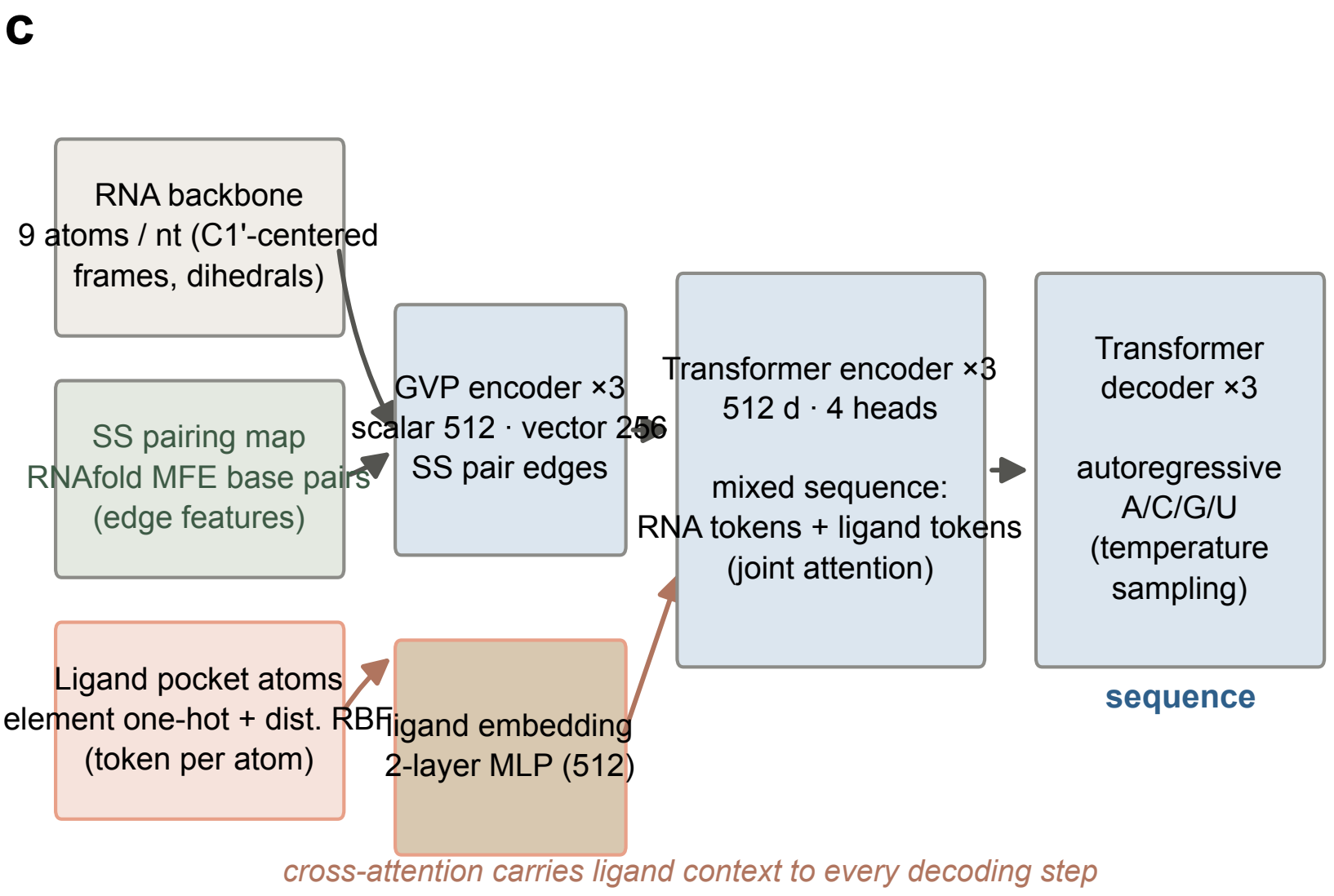
c
RNA backbone
9 atoms / nt (C1'-centered
frames, dihedrals)
SS pairing map
RNAfold MFE base pairs
(edge features)
Ligand pocket atoms
element one-hot + dist. RBF
(token per atom)
GVP encoder ×3
scalar 512 · vector 256
SS pair edges
ligand embedding
2-layer MLP (512)
Transformer encoder ×3
512 d · 4 heads
mixed sequence:
RNA tokens + ligand tokens
(joint attention)
Transformer
decoder ×3
autoregressive
A/C/G/U
(temperature
sampling)
sequence
cross-attention carries ligand context to every decoding step